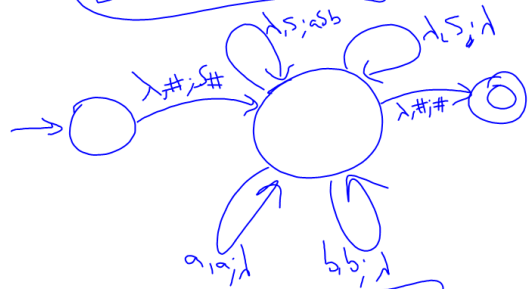


$L = \{a^n b^n\} \rightsquigarrow \text{APND}$

OPTION 1

$L \rightsquigarrow G1 \rightsquigarrow \text{APND}$

$S \rightarrow aSb \mid \lambda$



$\vdash a^n b^n =$

$$\begin{cases} S \rightarrow aBB/BAa/a \\ A \rightarrow ABa/BAa/a \\ B \rightarrow b/SA/abb \end{cases} \quad \text{if } \text{FNG?}$$
$$\left\{ \begin{array}{l} S \rightarrow aBb \mid BxAb \mid a \\ x_a \rightarrow a \\ A \rightarrow ABx_a \mid Bx_aA \mid a \\ B \rightarrow b \mid Sx_aB \mid ax_bxb \\ x_b \rightarrow b \end{array} \right.$$
$$A \rightarrow a x_1 x_2 \dots$$
$$\begin{aligned} A_0 &\rightarrow a_1 a_2 a_3 / a_4 a_5 a_6 / a \\ A_2 &\rightarrow a \\ A_3 &\rightarrow A_0 a_1 a_2 / a_3 a_4 a_5 / a \\ A_4 &\rightarrow b / A_1 a_2 a_3 / a_4 a_5 \\ A_5 &\rightarrow b \end{aligned}$$
$$A_1 \rightarrow aA_2A_3 / A_2A_3A_4/a$$
 ~~$A_2 \rightarrow a$~~
$$A_2 \rightarrow a$$

$$A_3 \rightarrow a / A_1 A_2 A_3 / A_1' / A_1 A_2 A_3 A_1'$$
$$f_3 \rightarrow \frac{A_1 A_2}{A_2 A_3}$$
$$A_4 \rightarrow b \mid a$$

$A_9 \rightarrow b$

$$\begin{aligned} \text{in} &\rightarrow A_2 A_4 A_2 A_4 / A_2 A_4 A_2 A_4 \end{aligned}$$
$$x \rightarrow b$$
$$A_3 \rightarrow a | b A_2 A_3 | a A_5 A_5 A_2 A_3 \dots$$